

CLIP Cross-Modal Image Pairing for 3D Scene Understanding

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Problem Statement

- Reconstructing 3D scenes from unstructured image collections is a core challenge in computer vision.
- Real-world scenarios introduce challenges:
 - Distractor images
 - Repetitive textures
 - Ambiguous scene boundaries
 - Occlusions or low-texture regions

Key Limitation

- Classical pipelines **struggle to identify which images belong to the same scene**, especially in cluttered environments.

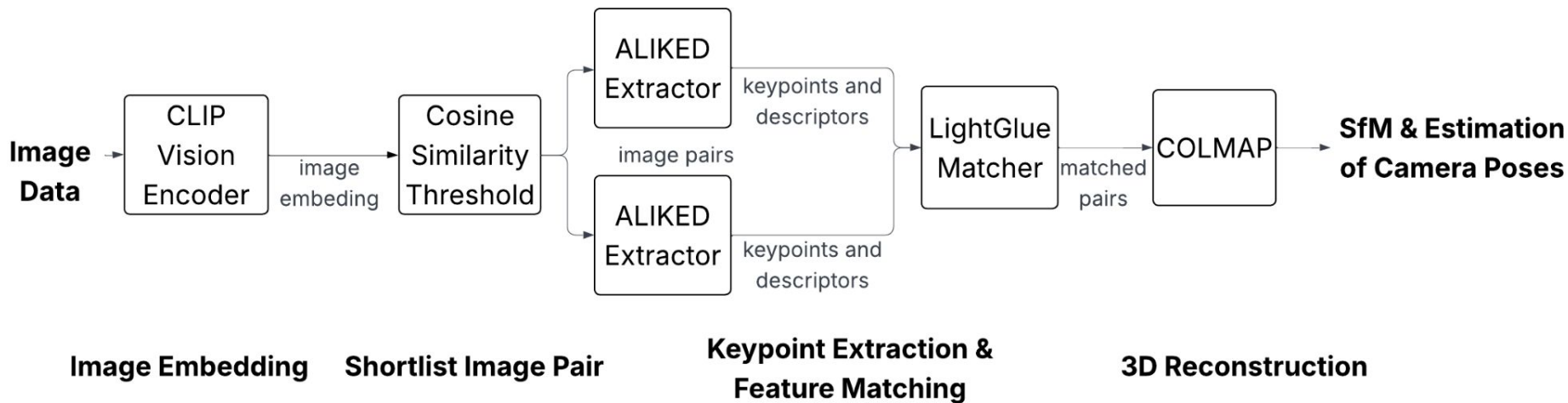
Motivation

- Leverage CLIP, a vision-language model pre-trained on large-scale image-text data.
- Use CLIP's semantic representations to:
 - Improve image clustering by grouping semantically similar content
 - Filter distractors more effectively than low-level features
 - Guide pair selection for downstream 3D reconstruction

Our Proposal

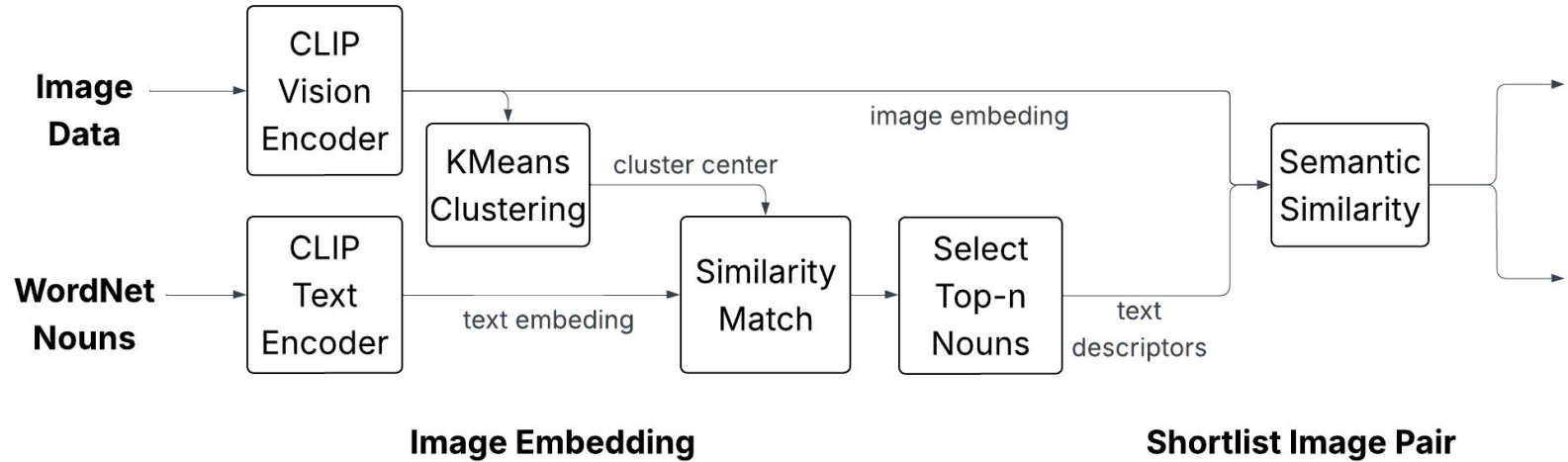
- CLIP Vision pipeline: Clusters images using visual semantic embeddings.
- CLIP Cross-Modal pipeline: Incorporates WordNet-derived text labels to enhance clustering and pair matching via multimodal guidance.

CLIP Vision Pipeline



*Baseline and pipeline come from a top-ranking method from Kaggle Image Matching Challenge 2024

CLIP Cross-Modal

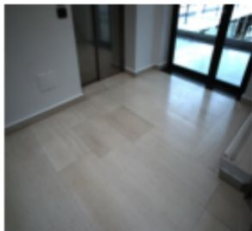


CLIP Text Counterpart Example: stairs dataset

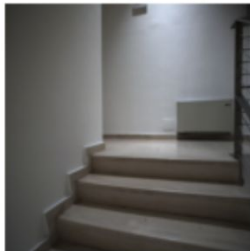
duplex_apartment
(stairs_split_1)



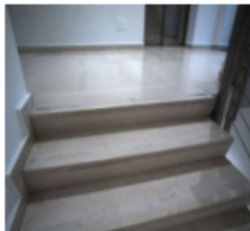
floor_board
(stairs_split_2)



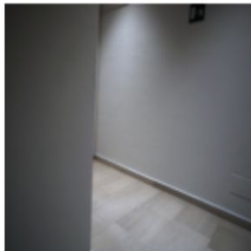
stairwell
(stairs_split_1)



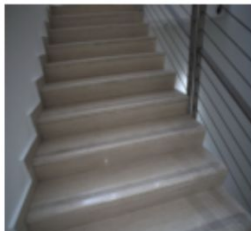
mezzanine_floor
(stairs_split_2)



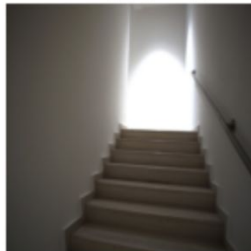
duplex_apartment
(stairs_split_1)



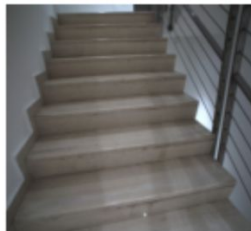
stair-carpet
(stairs_split_2)



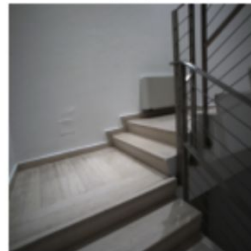
stairwell
(stairs_split_1)



floor_board
(stairs_split_2)



stairwell
(stairs_split_1)



mezzanine_floor
(stairs_split_2)



Comparison With Benchmark

Dataset	DINOv2		CLIP Vision		CLIP Cross-Modal	
	mAA	C	mAA	C	mAA	C
imc2023_haiper	63.89	63.53	62.78	63.53	65.56	63.53
imc2023_heritage	27.52	100.00	34.89	98.23	26.26	98.15
imc2023_theather_imc2024_church	37.14	100.00	40.00	100.00	35.71	100.00
imc2024_dioscuri_baalshamin	48.84	89.41	50.46	100.00	51.62	100.00
imc2024_lizard_pond	45.26	100.00	46.26	99.06	56.03	100.00
pt_brandenburg_british_buckingham	74.65	33.33	76.50	59.89	76.97	33.33
pt_piazzasanmarco_grandplace	80.25	50.00	79.48	100.00	80.71	100.00
pt_sacrecoeur_trevi_tajmahal	84.72	33.33	83.56	100.00	85.07	100.00
pt_stpeters_stpauls	78.99	50.00	78.61	99.50	78.61	50.00
amy_gardens	14.47	100.00	12.82	100.00	14.72	100.00
fbk_vineyard	20.62	33.33	9.58	33.33	9.74	33.33
ETs	28.85	43.18	23.08	45.24	28.85	43.18
stairs	2.22	71.43	2.78	77.14	2.78	50.00
Average	46.72	66.73	46.22	82.76	47.12	74.73

mAA(%): mean average accuracy

C(%): clusterness